

ECEN 743: Reinforcement Learning

Markov Decision Process (MDP)

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References

- [SB, Chapter 3-4]
 - [DB, Chapter 2]
 - [MMT, Chapter 6]
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- **Acknowledgment:** Some figures for this lecture are taken from UC Berkeley CS188 course, with permission.

Markov Decision Process (MDP)

Definition (Markov Decision Process (MDP))

A (discounted) Markov Decision Process (MDP) is a tuple $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$, where,

- $\mathcal{S}$ is a set of states (state space)
- $\mathcal{A}$ is a set of actions (action space)
- P is a transition probability function, $P(s'|s, a) = \mathbb{P}(s_{t+1} = s' | s_t = s, a_t = a), \forall s, s' \in \mathcal{S}, a \in \mathcal{A}$
- r is a reward function, $r(s, a), \forall s \in \mathcal{S}, a \in \mathcal{A}$
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 - A **stationary** policy $\pi : \mathcal{S} \rightarrow \Delta(\mathcal{A})$ specifies the control action for each state

Policy Evaluation

MDP Questions

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- How do we compute the value of a policy π ?

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- How do we compute the optimal policy?

$$\pi^* = \arg \max_{\pi \in \Pi} V_{\pi}$$

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- V_{π} is also called the **value function** of policy π

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- We will use the vector form V_{π}, r_{π} and the matrix form P_{π}

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- Intuition

$$V_\pi(s) = \underbrace{r_\pi(s)}_{\text{Expected reward for state } s \text{ under policy } \pi} + \gamma \underbrace{\sum_{s'} P_\pi(s, s') V_\pi(s')}_{\text{Expected value of the next state under policy } \pi}$$

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Computing V_π : Linear System of Equations Approach

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We can show that $(I - \gamma P_\pi)$ is invertible as follows.

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Let $V \in \mathbb{R}^{|\mathcal{S}|}$ be an arbitrary non-zero vector.

$$\begin{aligned}\|(I - \gamma P_\pi)V\|_\infty &= \|V - \gamma P_\pi V\|_\infty \stackrel{(a)}{\geq} \|V\|_\infty - \gamma \|P_\pi V\|_\infty \\ &\stackrel{(b)}{\geq} \|V\|_\infty - \gamma \|V\|_\infty = (1 - \gamma)\|V\|_\infty > 0\end{aligned}$$

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So, $(I - \gamma P_\pi)$ does not have 0 as eigenvalue. □

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- Policy evaluation using the linear system of equations approach has $O(|\mathcal{S}|^3)$ complexity

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Let $T_\pi V = r_\pi + \gamma P_\pi V$. Define the policy evaluation iteration $V_{k+1} = T_\pi V_k$. Then $\lim_{k \rightarrow \infty} V_k = V_\pi$

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 - ▶ $O(|S|^2)$
- How many iterations until convergence?

Action-Value (Q-value) function

- Recall that the value function of policy π is defined as

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$$Q_{\pi}(s, a) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) | s_0 = s, a_0 = a\right]$$

- It is straight forward to show that

$$Q_{\pi}(s, a) = r(s, a) + \gamma \sum_{s'} P(s' | s, a) V_{\pi}(s')$$

Action-Value (Q-value) function

- Recall that the value function of policy π is defined as

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- It is straight forward to show that

$$Q_{\pi}(s, a) = r(s, a) + \gamma \sum_{s'} P(s'|s, a) V_{\pi}(s')$$
$$Q_{\pi}(s, a) = \underbrace{r(s, a)}_{\text{Instantaneous reward for } (s, a)} + \gamma \underbrace{\sum_{s'} P(s'|s, a) V_{\pi}(s')}_{\text{Expected value of the next state}}$$

Action-Value (Q-value) function

- Recall that the value function of policy π is defined as

$$V_{\pi}(s) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 = s\right]$$

- Action-Value function (Q-value function) for policy π is defined as

$$Q_{\pi}(s, a) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 = s, a_0 = a\right]$$

- It is straight forward to show that

$$\begin{aligned} Q_{\pi}(s, a) &= r(s, a) + \gamma \sum_{s'} P(s' \mid s, a) V_{\pi}(s') \\ Q_{\pi}(s, a) &= \underbrace{r(s, a)}_{\text{Instantaneous reward for } (s, a)} + \gamma \underbrace{\sum_{s'} P(s' \mid s, a) V_{\pi}(s')}_{\text{Expected value of the next state}} \end{aligned}$$

- It is also straight forward to show that

$$V_{\pi} = \sum_a Q_{\pi}(s, a) \pi(s, a)$$